

8-WEEK DAILY BUILD PLAN

RaktSetu — Real-Time Blood Donation & Blood Bank Ecosystem · Pilot-Ready MVP

Akhilan & Karthikeyan · Skill-aligned task split · Pilot-city launch · Next.js 15 PWA frontend

Why this split

Akhilan is a final-year CS student at VIT — strong fundamentals, fast learner, first production system. Karthikeyan is an experienced fullstack developer with UI/UX depth. The realistic split:

- Akhilan owns: Data store, backend API, geo search (PostGIS), wave-dispatch engine, notification pipeline, eligibility/matching logic, test coverage. (Highest learning curve — every backend PR reviewed by Karthikeyan.)
- Karthikeyan owns: Entire frontend (Next.js 15 PWA + shadcn/ui), all UI/UX design decisions, Docker/Compose builds, CI/CD, production deploy, monitoring.
- Both pair on: Week 1 setup, the Week 5 wave-dispatch integration (the make-or-break week), weekly Friday integration check, pilot launch.
- Mentorship rule baked in: one shared TypeScript language across the whole stack (Node backend, Next.js frontend) so Karthikeyan can meaningfully review every line Akhilan writes. No solo merges to main after CI exists.

Stack rationale (single-language TypeScript, PWA-first)

- PWA, not native apps: two people, 8 weeks. A Next.js 15 installable PWA (manifest + service worker + web push) ships to Android/iOS/desktop from one codebase. Native apps deferred to v1.1.
- Node 20 + Express + Prisma (not FastAPI): keeps backend in TypeScript so the senior dev can review the student's code without context-switching. Prisma for typed queries + migrations; raw SQL only for PostGIS.
- PostgreSQL 16 + PostGIS: radius search ('O— donors within 10 km') is THE core query. ST_DWithin on a geography column with a GiST index — not Haversine math in app code.
- Redis 7 + BullMQ: wave dispatch is queue-driven — delayed jobs implement 'Wave 2 fires in 5 minutes unless someone accepts'. Gentle production-jobs pattern for a student to learn.
- shadcn/ui + Tailwind: Radix-based accessible components (Dialog, Sheet, InputOTP, Form) copied into the repo. Karthikeyan's UX effort goes into the emergency flow, not rebuilding a button.
- Auth.js v5 Credentials → backend OTP: backend is source of truth for JWT. MSG91 sends OTP SMS (India DLT template approval takes days — submitted Day 1).
- FCM web push + MSG91 SMS fallback: emergency notifications must arrive. Push first; undelivered after 90 s → SMS. Both logged with delivery receipts.
- Google Maps Platform: geocoding + navigation deep links. No in-app turn-by-turn needed for MVP.

MVP scope reduction (so 8 weeks is realistic)

The full PRD v1.0 is too much for two people in 8 weeks. Cut to pilot scope:

- KEEP for MVP: OTP auth · Donor registration + verification tiers 0–1 · Live availability + eligibility engine · PostGIS search with compatibility expansion · One-tap emergency request · Wave-based dispatch (rules-ranked) · FCM push + SMS fallback · Accept → contact reveal → navigation · Post-accept chat · Donation logging + PDF certificates with QR verify · Hospital mini-portal · Admin console · DPDP basics.
- DEFER to v1.1: Native apps · Masked proxy calls (MVP reveals number only post-accept) · Blood bank component/expiry module · ML ranking · AI chat assistant · Demand prediction · e-RaktKosh integration · DigiLocker ID · Multi-language · NGO camp module.
- Rationale: the emergency loop (request → waves → accept → arrive → donate) IS the product. Everything else is decoration until that loop works flawlessly in one pilot city.

Differentiators vs existing donor-directory apps (baked into the daily plan)

- FR-EMG-01 — Wave dispatch, never broadcast (Week 5, Days 21–25): ranked batches of 10 with timed escalation. Directory apps blast everyone → notification fatigue → dead platform.
- FR-DON-04 — Live availability engine (Week 3, Day 11): eligibility countdown + snooze + inactivity decay. Directory apps show donors who donated last week.
- FR-EMG-03 — Inventory-first flow (Week 7, Day 33): hospital stock surfaced before donors are disturbed. Directory apps cannot see stock at all.
- FR-PRV-01 — Zero phone-number exposure pre-accept (Week 6): contact revealed only after mutual match. Directory apps print raw numbers publicly.

- FR-DON-06 — Verified-donation certificates with public QR verify URL (Week 7, Day 32): a trust artifact directory apps don't have.
- FR-DAT-01 — Stale-profile decay + OTP revalidation (Week 3, Day 14): the #1 complaint about existing apps is dead data; this platform garbage-collects itself.

Stack at a glance

Layer	Technology	Owner
Backend API	Node.js 20 + Express + TypeScript + zod	Akhilan
ORM / Migrations	Prisma (raw SQL for PostGIS queries)	Akhilan
Database	PostgreSQL 16 + PostGIS (GiST geo index)	Akhilan
Cache / Queue	Redis 7 + BullMQ (delayed jobs = waves)	Akhilan
Object Storage	MinIO (certificates, verification docs)	Akhilan
Auth (backend)	OTP via MSG91 → JWT (jose) + RBAC	Akhilan
Push / SMS	FCM (web push) + MSG91 SMS fallback	Akhilan
Geo / Maps APIs	Google Maps Platform (geocode, deep links)	Akhilan
PDF Certificates	pdfkit + qrcode	Akhilan
Frontend Framework	Next.js 15 (App Router) + React 18 + TypeScript	Karthikeyan
UI Components	shadcn/ui (Radix primitives + Tailwind)	Karthikeyan
Frontend Auth	Auth.js v5 Credentials → backend /auth/verify-otp	Karthikeyan
Data Fetching	TanStack Query (polling for live request state)	Karthikeyan
Forms / Validation	react-hook-form + zod (schemas shared with backend)	Karthikeyan
PWA	Service worker · manifest · web push · offline queue	Karthikeyan
Icons / Font / Toasts	lucide-react · Inter via next/font · sonner	Karthikeyan
Charts (admin)	Recharts	Karthikeyan
Containers	Docker Compose (dev) → compose-prod with Caddy	Karthikeyan
CI / CD	GitHub Actions (lint, typecheck, vitest, build)	Karthikeyan
Monitoring	Sentry (both SDKs) + UptimeRobot + pg_dump cron	Karthikeyan
Hosting	Hetzner/DO VPS (Mumbai) + Caddy auto-SSL + Cloudflare	Karthikeyan

MONTH 1 — Foundation, Donors & Requests · Weeks 1–4

End goal: verified donors registered, searchable by blood group + radius with live availability. Emergency requests creatable. No dispatch yet.

WEEK 1 — Shared setup & Data Store

Both pair daily this week. Also: submit MSG91 DLT template approval Day 1 — it takes days and Week 5 depends on it.

Day	Akhilan — Backend / Data / Dispatch	Karthikeyan — Frontend / UX / Deploy	Done When
Day 1	- Install Node 20 LTS, Docker Desktop, VS Code, git - Create GitHub repo with folders: backend/ frontend/ infra/ + README + .gitignore (Node, .env, .next/) - Run: docker run hello-world (confirm Docker works) - Submit MSG91 signup + DLT OTP/alert template registration (approval takes 2–5 days — start NOW)	- Verify Node 20 · clone the repo Akhilan just created - Inside frontend/: npx create-next-app@latest raktsetu-ui --typescript --tailwind --app --src-dir --eslint - Verify: npm run dev → http://localhost:3000 - Sketch the 3 core flows on paper: donor onboarding, emergency request, donor accept (UX is his call)	- Both run hello-world - Next.js dev server loads at :3000 - DLT registration submitted
Day 2	- Write docker-compose.yml — 3 services: postgres/postgis:16 (5432), redis:7-alpine (6379), minio/minio (9000, 9001) - Run: docker compose up -d → confirm all 3 RUNNING - Write .env.example: DATABASE_URL, REDIS_URL, JWT_SECRET, MSG91_KEY, FCM creds, MINIO creds - Add .env to .gitignore (NEVER commit .env)	- Self-host Inter via next/font/google · npm i lucide-react - npx shadcn@latest init (Slate, CSS variables) · add: button card input label dialog sheet form toast sonner badge input-otp progress - Theme tokens: primary red-600 (blood context), success emerald, warning amber - Scaffold routes: app/(auth)/login · app/(app)/{home,request,search,requests/[id],profile,history} · app/(portal)/hospital · app/(admin)	- docker compose ps shows 3 green .env.example committed - Theme renders, routes scaffolded
Day 3	- prisma init · write schema: User, DonorProfile, Institution, BloodRequest, Match, Donation, NotificationLog, AuditLog - DonorProfile gets location geography(Point,4326) via raw SQL migration + GiST index - prisma migrate dev · verify tables in psql (\dt)	- Build app shell: mobile-first bottom tab bar (Home · Search · SOS center button · History · Profile) + desktop sidebar at md: - SOS button: oversized, red, center-docked — the emergency entry point (auth) layout minimal, no shell	8 tables in Postgres - GiST index confirmed via \d - Shell renders at both breakpoints
Day 4	- Enable postgis extension in migration - Seed script: 1 admin, 1 hospital institution, 20 fake donors scattered around pilot city with realistic coordinates + varied blood groups - Verify: raw ST_DWithin query returns nearby donors	- Build LoginForm: phone input (+91 mask) → 'Send OTP' → shadcn InputOTP 6-digit screen with 60 s resend cooldown (UI only) - Home placeholder: availability card, eligibility ring, stats — all '—'	- Seed inserts 20 geo-scattered donors - Radius SQL query works - OTP UI flows end to end
Day 5	- Shared zod schema package packages/shared (BloodGroup enum, compatibility matrix constant, request/donor DTOs) — imported by both apps - SCHEMA.md with mermaid ER diagram · pre-commit: eslint + prettier - Tag: v0.0.1-data-store-ready	- Review schema with Akhilan — confirm every field the UI needs exists (esp. Match timestamps for the tracking stepper) - STRUCTURE.md · scripts: lint / format / typecheck · pre-commit hooks - Tag: v0.0.1-frontend-skeleton	- Schema reviewed - Shared package imported by both apps - Tags pushed

WEEK 2 — OTP auth + API skeleton + donor registration

Day	Akhilan — Backend / Data / Dispatch	Karthikeyan — Frontend / UX / Deploy	Done When
Day 6	- Express app: GET /health → {status, db_ok, redis_ok} · CORS allow :3000 - Error middleware (zod errors → 422, unknown → 500 + log) - Add backend + worker services to compose (hot reload via tsx watch)	- Install Auth.js v5: npm i next-auth@beta - CredentialsProvider authorize() POSTs phone+otp to backend /auth/verify-otp, returns JWT + user - JWT session strategy · typed fetch wrapper lib/api.ts attaching Bearer token	- GET /health green from browser /api/auth/providers responds

Day	Akhilan — Backend / Data / Dispatch	Karthikeyan — Frontend / UX / Deploy	Done When
Day 7	- POST /auth/request-otp: 6-digit code in Redis otp:{phone} EX 300, send via MSG91 (dev mode: console log) - Rate limit: 3 OTP/phone/hour, 10/IP/hour - POST /auth/verify-otp → JWT (8 h) + refresh token · GET /auth/me	- Wire login E2E: phone → OTP → signIn('credentials') → /home - sonner toast on wrong OTP with attempts-left - middleware.ts protects (app), (portal), (admin) routes · 'Remember me' → 30 d session	- Real login works with console OTP - Middleware redirects correctly - Rate limit returns 429
Day 8	- RBAC roles: donor · recipient · hospital_staff · coordinator · admin (a user can hold several) - requireRole() middleware - Every mutating endpoint writes AuditLog {userId, action, entity, ts}	- Role-aware navigation (hospital_staff sees portal link, admin sees admin) - Profile page skeleton: identity card, roles, consent toggles (location, notifications)	- Wrong role gets 403 on admin route - Audit rows written
Day 9	- BullMQ: queue 'dispatch', worker process, first dummy job (sleep 3 s, log) - Worker as separate compose service - Repeatable-job scaffold (used later for decay + revalidation crons)	Build donor registration wizard (the UX centerpiece): Step 1 basics (name, DOB, sex, weight) → Step 2 blood group (8 big chips + 'not sure') → Step 3 health questionnaire (12 yes/no, one per screen, thumb-reachable) → Step 4 location consent + GPS capture	- Worker picks up job - Wizard navigates all steps with validation
Day 10	- POST /donors: creates DonorProfile, stores geo point, computes initial eligibility from questionnaire (age 18–65, weight ≥ 50 kg, deferral answers) - GET /donors/me · PATCH /donors/me · consent records persisted per DPDP	- Wire wizard → API, optimistic success screen ('You're donor #21 in your city') - Home shows real availability card + blood group - Tag: v0.1.0-auth-donors	- Registration E2E creates geo-located donor - Ineligible answers set correct deferral - Tag pushed

WEEK 3 — Availability engine + geo search + verification

Day	Akhilan — Backend / Data / Dispatch	Karthikeyan — Frontend / UX / Deploy	Done When
Day 11	- FR-DON-04 availability engine: states AVAILABLE / UNAVAILABLE / SNOOZED(until) / INELIGIBLE(auto) / UNCONFIRMED(auto) - Transition rules as one pure function + unit tests - Eligibility countdown: last donation + 90 d (male) / 120 d (female), configurable - PATCH /donors/me/availability	- Availability UI on Home: large switch + snooze picker (1 w / 2 w / 1 mo / custom) - Eligibility countdown ring (SVG, days remaining) — grays out switch while INELIGIBLE with explainer - Status chip color system used everywhere	- State machine tests green - Toggle + snooze persist - Countdown accurate for seeded donors
Day 12	- Geo search: GET /donors/search?bg=&lat=&lng=&radiusKm= — raw SQL ST_DWithin + GiST, filters AVAILABLE + eligible only - Compatibility expansion from shared matrix (request B- → return B-, O-) with exact-match ranked first (protect O-) - Return FUZZED coordinates (~1 km jitter) + distance band ('2–3 km'), never exact location	- Search page: blood group chip selector, radius slider (5–50 km), results as donor cards (initials avatar, blood group, distance band, badges, availability chip) - Empty state: 'No available donors — widen radius or raise a request' with CTA to SOS	- Search returns correct compat-expanded set - Exact-group sorts first - Only fuzzed coords in payload (network tab check)
Day 13	- Verification tiers: Tier 0 = OTP phone (default) · Tier 1 = document upload (MinIO presigned PUT) for blood-group proof, admin approves - POST /donors/me/documents · admin review endpoint	- VerificationBadge component (shield icons: gray/blue/green with tooltip explaining what was verified) - Upload flow in Profile (camera/file, preview, submit) - Map view toggle on search: clustered fuzzed markers	- Upload → admin approve → badge upgrades - Map shows fuzzed cluster, not exact pins
Day 14	- FR-DAT-01 data hygiene crons (BullMQ repeatable): nightly stale-decay — no activity 60 d → UNCONFIRMED + re-verification push queued; monthly OTP revalidation prompt - Dedup guard on phone + device fingerprint at registration	- Re-verification prompt UX: one-tap 'Still available?' notification → deep link → single confirm button (renewal must be effortless — this is why data stays fresh) - Profile completeness meter nudging Tier 1	- Cron flips a stale seed donor to UNCONFIRMED - One-tap reconfirm restores AVAILABLE

Day	Akhilan — Backend / Data / Dispatch	Karthikeyan — Frontend / UX / Deploy	Done When
Day 15	- Vitest: compatibility matrix (all 8×8 cases), eligibility calculator edges, search filter correctness - Bruno/Postman collection for all endpoints - Tag: v0.1.0-search-ready	- Loading skeletons, error toasts on all data screens - Mobile pass at 375 px · large-touch check on availability + search - Tag: v0.1.0-ui-week3	- Tests green (matrix 100% covered) - UX clean on phone - Tags pushed

WEEK 4 — Emergency requests + institution directory · Month 1 gate

Day	Akhilan — Backend / Data / Dispatch	Karthikeyan — Frontend / UX / Deploy	Done When
Day 16	- BloodRequest: POST /requests (bg, units, institution or free-text location, urgency CRITICAL/URGENT/PLANNED, patient note) - State machine: CREATED → SEARCHING → MATCHED → FULFILLED / EXPIRED / CANCELLED as pure function + tests - AuditLog every transition	- SOS flow (the flagship UX): tap SOS → 3 inputs max (blood group, units stepper, hospital picker with search) → urgency → confirm - Target: raise a request in under 60 s, one-handed, under stress · oversized controls, no optional fields on this path	- Request created < 60 s in usability self-test - Invalid transitions rejected - Audit trail complete
Day 17	- Institution directory: import pilot-city hospital + blood bank list (public e-RaktKosh directory CSV) with coordinates - GET /institutions/nearby?lat=&lng=&type= (PostGIS again)	- Nearby hospitals/blood banks screen: list + map, each card with 'Call' (tel:) and 'Directions' (Maps deep link) - Works logged-out too (public route) — emergency info shouldn't need an account	30+ pilot-city institutions seeded with coords - Deep links open dialer + Maps
Day 18	- Anti-abuse: request rate limit (3 open/account), duplicate detection (same bg + institution + overlapping window → warn/merge) - Coordinator review flag on suspicious patterns	Live request tracking screen (recipient): vertical stepper of states with timestamps, 'donors notified / responded / accepted' counters (mock until Week 5), cancel with reason dialog	- Duplicate request warns - Tracking stepper renders all states from fixtures
Day 19	- GET /requests/me history + detail · close/cancel with reason enum (fulfilled-elsewhere, patient-recovered...) — feeds analytics - Expiry job: auto-EXPIRED after urgency-based TTL (6 h critical / 24 h urgent / 7 d planned)	- History screen: request cards with outcome chips · closure dialog (reason select + note) - Recipient home: active request pinned to top with live status chip	- Expiry job flips stale request - Closure reasons persisted
Day 20	- Month-1 bug bash (both): full pass as donor, recipient, admin; attempt auth bypasses + direct-ID access to others' requests/profiles → must 403 - Fix criticals same day · Tag: v0.2.0-month1-complete	- Same bash from UX side: cross-browser (Chrome/Firefox/Safari), 375 px pass on every screen, empty/error states everywhere - Tag: v0.2.0-month1-frontend	- Zero cross-user data leaks - Every screen has empty + error state - Tags pushed

MONTH 2 — Dispatch, Match Lifecycle & Pilot Launch · Weeks 5–8

End goal: full emergency loop live in production for one pilot city: request → waves → accept → navigate → donate → certificate. Hospital mini-portal + admin console. Pilot partner onboarded.

WEEK 5 — Wave dispatch + notifications (the make-or-break week — pair daily)

Day	Akhilan — Backend / Data / Dispatch	Karthikeyan — Frontend / UX / Deploy	Done When
Day 21	- FCM Admin SDK: POST /devices registers push token per user/device - sendPush(userId, payload) service with delivery-receipt logging to NotificationLog - Topic-free, direct-token sends only	- Web push in PWA: permission prompt asked in context (after donor registration, not first visit) - Service worker push + notificationclick handlers deep-linking to the request - In-app notification inbox screen	- Push arrives on locked Android phone - Tap opens the right request - Receipts logged

Day	Akhilan — Backend / Data / Dispatch	Karthikeyan — Frontend / UX / Deploy	Done When
Day 22	- Ranking formula (rules now, ML later): $\text{score} = w1 \cdot \text{exactMatch} + w2 \cdot \text{proximity} + w3 \cdot \text{eligibilityMargin} + w4 \cdot \text{reliability} + w5 \cdot \text{verificationTier}$ — pure function + tests, weights in config <code>rankDonors(requestId)</code> returns ordered candidate list	- Donor-side request card: full-screen actionable view — blood group (huge), hospital, distance band, urgency, patient note - Two buttons only: Accept / Can't this time (decline reasons: too far, unwell, busy — feeds reliability fairly)	- Ranking deterministic + tested - Accept/decline round-trips to API
Day 23	- FR-EMG-01 wave engine (BullMQ delayed jobs): Wave 1 = top 10 → no accept in 5 min → Wave 2 = next 15 + radius +5 km → 10 min → Wave 3 + ESCALATED flag - First accept CANCELS pending wave jobs + expires other donors' pending cards - Match rows: NOTIFIED → RESPONDED → ACCEPTED timestamps	- Recipient live tracking goes real: TanStack Query 5 s polling while SEARCHING — animated counters (notified/responded/accepted), wave indicator ('Wave 2 · searching wider') - On accept: celebratory state + donor first-name/initial + ETA card	- Full wave cascade observable in logs - Accept cancels remaining jobs - Tracking updates < 5 s
Day 24	- SMS fallback: push undelivered after 90 s → MSG91 SMS (DLT template) with short link - Quiet hours + fatigue cap: max 4 emergency pings/donor/month unless CRITICAL, config-backed - All sends in NotificationLog with channel + status	- Post-accept contact + navigation: contact card reveals recipient phone to donor (and vice-versa) ONLY now — before this, zero personal data crossed 'Navigate' button → Google Maps deep link to hospital · recipient sees 'Donor en route'	- SMS lands on real phone - Fatigue cap blocks 5th ping - No phone number in any pre-accept payload (verified)
Day 25	- E2E: seed 20 donors → raise request → Wave 1 push → decline all → Wave 2 fires at 5 min → accept → jobs cancelled → contact revealed - Chaos: restart worker mid-wave, jobs resume (BullMQ persistence) - Tag: v0.3.0-dispatch-ready	- Same E2E from 3 physical phones (2 donors, 1 recipient) - Fix every UX hitch found (notification copy, button sizes, state confusion) - Tag: v0.3.0-dispatch-ui	- Full loop works across real devices - Worker restart loses nothing - Tags pushed

WEEK 6 — Match lifecycle, chat, reliability, escalation

Day	Akhilan — Backend / Data / Dispatch	Karthikeyan — Frontend / UX / Deploy	Done When
Day 26	- Extend Match machine: ACCEPTED → ARRIVED → DONATED / NO_SHOW / WITHDRAWN with transition endpoints + who-may-transition rules (donor marks arrived; recipient or hospital confirms donated) - Withdrawal re-opens dispatch (resume next wave)	- Donor journey screens per state: en-route ('I've arrived' button), arrived (waiting confirm), donated (celebration + certificate teaser) - Recipient mirror states · withdraw flow with confirm dialog ('This restarts the donor search')	- Every transition enforced server-side - Withdraw resumes waves - Both sides see consistent state
Day 27	- Post-accept in-app chat: REST + 3 s polling (Socket.io deferred — polling is fine at pilot scale), messages scoped to Match, closed when Match closes - Content guard: strip phone/UPI patterns in pre-accept contexts (defense in depth)	- Chat UI inside match screen: bubbles + quick-reply chips ('On my way', 'ETA 20 min', 'Which ward?', 'Reached gate') — donors won't type while traveling - Unread badge on tab	- Chat round-trips between two phones - Quick replies one-tap - Chat locks on match close
Day 28	- Reliability score nightly job: <code>acceptRate</code>, <code>showUpRate</code>, cancellations, profile freshness → 0–100 on <code>DonorProfile</code>, feeds ranking w4 - NO_SHOW applies cooldown (excluded from waves 14 d) with appeal flag	- Donor profile: reliability with supportive framing ('Trusted responder' tiers, not a shaming number) - Milestone badges UI (First Donation, 3 Lives, 5 Lives...) · history timeline	- Score recomputes nightly on seeds - No-show cooldown excludes donor from next dispatch
Day 29	- Escalation: Wave 3 exhausted → request flagged to coordinator queue GET /coordinator/queue · assign / handle / resolve endpoints - Emergency broadcast endpoint (coordinator-triggered, opted-in donors only, city-wide, hard-capped)	- Coordinator mini-console (desktop): escalated requests sorted by urgency + age, assign-to-me, resolution notes - Broadcast button with are-you-sure + reach estimate	- Exhausted request appears in queue < 1 min - Broadcast reaches only opted-in donors

Day	Akhilan — Backend / Data / Dispatch	Karthikeyan — Frontend / UX / Deploy	Done When
Day 30	- Load test: 50 concurrent requests × 20-donor pool — queue depth, slow query log - Add indexes: (bg, status), Match(request_id); verify geography GIST - Tag: v0.4.0-lifecycle	- Full-flow demo recording (screen capture) for pilot-partner pitch - Polish pass on all Match screens - Tag: v0.4.0-lifecycle-ui	- P95 dispatch enqueue < 15 s under load - Demo video cut - Tags pushed

WEEK 7 — Donations, certificates, hospital portal, admin, privacy

Day	Akhilan — Backend / Data / Dispatch	Karthikeyan — Frontend / UX / Deploy	Done When
Day 31	- Donation logging: hospital-confirmed (via portal, verified=true) vs donor self-report (verified=false chip) - Completion auto-starts eligibility countdown + flips availability to INELIGIBLE	- Donation history screen: entries with verified / self-reported chips - Countdown ring on Home driven by real donation data 'Log a past donation' flow (date + place) for onboarding existing donors	- Confirmed donation starts countdown - Self-report clearly distinguished
Day 32	- FR-DON-06 certificates: pdfkit certificate (name, date, institution, blood group) + QR → public GET /verify/{donationId} - Store in MinIO, presigned download URL	- Certificate UI: preview, download, native share sheet (Web Share API) - Public verify page (no auth): green verified card with donation facts — employers/colleges can check it	- PDF renders in Adobe + Chrome + phone - QR scan → verify page confirms
Day 33	- Hospital mini-portal APIs: staff onboarding (admin-approved) · verify/attach patient requests (FR-EMG-03: verified requests get badge + ranking boost) - Simple inventory (units per blood group, manual update) surfaced at request creation: 'City Blood Bank shows 2 units of B— call before notifying donors'	- Hospital portal (desktop): request verification queue · inventory grid (8 groups × unit stepper, single Save) · audit trail view - Inventory hint card injected into recipient SOS flow when stock exists nearby	- Staff verifies a request → badge visible to donors - Stock hint appears before dispatch
Day 34	- Admin console APIs: user search/block, fraud review queue (flagged requests, dedup hits), live request monitor, NotificationLog viewer - Metrics endpoint: requests today, fulfillment rate, median time-to-accept	- Admin pages: DataTables with block/unblock, fraud queue approve/reject, live request board - Metrics cards + Recharts sparkline (median time-to-accept trend — the north-star number)	- Admin blocks a user → sessions revoked - Metrics match DB reality
Day 35	- DPDP pass: consent records audited · GET /me/export (JSON bundle) · DELETE /me (soft-delete + PII scrub, retain anonymized audit) - Grep-level check: no exact coords or phone in any pre-accept response - Tag: v0.5.0-portal-privacy	- Settings: consent toggles, 'Download my data', delete-account flow (typed confirm + cooling-off notice) - Privacy explainer in plain language ('When is my number shared? Only after you accept.') - Tag: v0.5.0-ui	- Export bundle complete - Deletion scrubs PII, keeps stats - Privacy claims match observed payloads

WEEK 8 — Hardening, production deploy, pilot launch

Day	Akhilan — Backend / Data / Dispatch	Karthikeyan — Frontend / UX / Deploy	Done When
Day 36	- Security audit: OWASP top-10 pass, helmet, rate limits on every mutating route, npm audit clean, JWT expiry/refresh checked - Offline emergency queue: POST /requests accepts client-generated UUID (idempotent) so queued offline requests don't duplicate on sync	- PWA hardening: manifest + icons + install prompt · offline shell caches institution directory + emergency numbers (usable with zero signal) - Background sync queues an offline SOS → sends on reconnect - Lighthouse: Perf > 85, A11y > 90, PWA installable	- Offline SOS queues and sends exactly once on reconnect - Installs to home screen - Lighthouse targets hit
Day 37	- Vitest sweep: state machines, ranking, compat matrix, eligibility, wave cancellation — CI-gating suite green - Structured logging (pino JSON) across API + worker	- Dockerfiles: backend + worker (node:20-alpine multi-stage), frontend (Next standalone, < 250 MB) - docker-compose.prod.yml: healthchecks, restart policies, Caddy (app./api. domains) - GitHub Actions: lint + typecheck + vitest + build · branch protection on main	- CI green + gating - Prod compose runs locally - Solo merges now impossible

Day	Akhilan — Backend / Data / Dispatch	Karthikeyan — Frontend / UX / Deploy	Done When
Day 38	- Backup cron: nightly pg_dump → MinIO backups/, 30 d retention · test restore locally - Runbook: deploy, rollback, restore, rotate JWT secret, block abusive user, resume stuck queue	- VPS (Mumbai region) + domain + Cloudflare DNS · UFW 22/80/443, key-only SSH - Deploy prod compose, Caddy auto-SSL - securityheaders.com fixes via Caddy (HSTS, CSP with Next 15 nonces, XFO)	- App live on real domain, green padlock - Restore drill passed - Headers grade A
Day 39	- Sentry backend SDK · chaos on prod: kill worker mid-wave, restart Redis → dispatch resumes, no orphan matches - Real-SMS + real-push E2E on production with 3 phones	- Sentry Next.js wizard (client + server) · UptimeRobot on app + api /health, SMS alert on 2-min downtime - Full E2E on prod from the same 3 phones · fix launch blockers	- Test errors land in Sentry (both) - Prod E2E clean incl. real SMS - Uptime monitor live
Day 40	- Pilot launch prep: pilot-city institutions verified, coordinator account for NGO partner, fatigue caps + quiet hours confirmed in prod config 8-week retrospective doc: shipped / slipped / next - Tag: v1.0.0-pilot	- Pilot kit: 2-page onboarding PDF (screenshots), 5-min demo script rehearsed 3×, QR posters for the partner blood-camp/college drive - v1.1 backlog board (masked calls, native apps, blood-bank module, ML ranking, e-RaktKosh) - Tag: v1.0.0-pilot	- NGO/college partner onboarded with coordinator access - Demo runs cleanly 3× - v1.0.0 tagged

8-WEEK SUMMARY

Month	Phase	Week Range	Milestone
Month 1	Foundation, Donors & Requests	Weeks 1–4	OTP auth · geo-located verified donors with live availability + eligibility engine · PostGIS compat-expanded search · one-tap emergency requests with state machine · institution directory · data-hygiene crons. Tag v0.2.0.
Month 2	Dispatch, Lifecycle & Launch	Weeks 5–8	Wave dispatch (FR-EMG-01) with push + SMS fallback · full match lifecycle with chat + reliability · certificates with QR verify · hospital mini-portal with inventory-first hints (FR-EMG-03) · admin console · DPDP compliance · production deploy + monitoring · pilot city LIVE with NGO partner. Tag v1.0.0-pilot.

Rules for staying on schedule

- Never start a week until the previous week's tag is pushed. Week 5 dispatch WILL fail if Week 3's availability engine is incomplete.
- Demo to each other every Friday. If you can't demo it, it's not done.
- Cut scope before cutting quality. Defer the hospital portal before shipping a flaky dispatch loop — the emergency path is sacred.
- Karthikeyan reviews every Akhilan PR (mentorship + safety). Akhilan reviews Karthikeyan's for learning. Solo merges to main forbidden once CI exists (Day 37).
- One bug-bash day per week (Friday afternoon). Don't carry tech debt past 5 days.
- Real phones every week from Week 5. Push notifications and PWA behavior lie in devtools; only physical devices tell the truth.
- Pilot pull > feature push from Week 8. Build what the NGO partner and first donors ask for, not what looks impressive.

Tech-stack split summary

Akhilan: Node.js 20 · Express · TypeScript · zod · Prisma · PostgreSQL 16 + PostGIS · Redis 7 · BullMQ · MinIO · MSG91 (OTP + SMS) · FCM Admin SDK · Google Maps APIs · pdfkit + qrcode · pino · Vitest

Karthikeyan: Next.js 15 (App Router) · React 18 · TypeScript · Tailwind CSS · shadcn/ui · Auth.js v5 · TanStack Query · react-hook-form + zod · PWA (service worker, web push, background sync) · lucide-react · sonner · Recharts · Docker + Compose · GitHub Actions · Caddy · Sentry · UptimeRobot · Cloudflare

Both: [git](#) · [GitHub](#) · [GitHub Container Registry](#) · [Bruno/Postman](#) · [runbook](#) · [pilot onboarding kit](#)

END OF PLAN